Classification:

Generally metal carbonyls are classified in two different ways as described below.

(i) Classification based on the number of metal atoms present.

Depending upon the number of metal atoms present in a given metallic carbonyl, they are classified as follows.

a. Mononuclear carbonyls

These compounds contain only one metal atom, and have comparatively simple structures. For example, $\left[Ni(CO)_4\right]$ - nickel tetracarbonyl is tetrahedral, $\left[Fe(CO)_5\right]$ - Iron pentacarbonyl is trigonalbipyramidal, and $\left[Cr(CO)_6\right]$ - Chromium hexacarbonyl is octahedral.

b. Poly nuclear carbonyls

Metallic carbonyls containing two or more metal atoms are called poly nuclear carbonyls. Poly nuclear metal carbonyls may be Homonuclear $\left(\left[Co_2(CO)_8\right], \left[Mn_2(CO)_{10}\right], \left[Fe_3(CO)_{12}\right]\right)$ or hetero nuclear $\left(\left[MnCo(CO)_9\right], \left[MnRe(CO)_{10}\right]\right)$ etc.

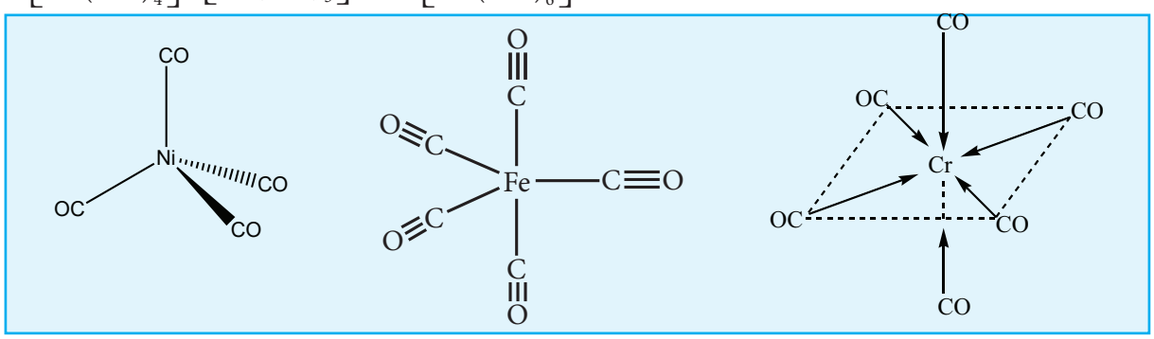
(ii) Classification based on the structure:

The structures of the binuclear metal carbonyls involve either metal–metal bonds or bridging CO groups, or both. The carbonyl ligands that are attached to only one metal atom are referred to as **terminal** carbonyl groups, whereas those attached to two metal atoms simultaneously are called *bridging* carbonyls. Depending upon the structures, metal carbonyls are classified as follows.

a. Non-bridged metal carbonyls:

These metal carbonyls do not contain any bridging carbonyl ligands. They may be of two types.

(i) Non- bridged metal carbonyls which contain only terminal carbonyls. Examples: $\left[Ni(CO)_4\right]$, $\left[Fe(CO)_5\right]$ and $\left[Cr(CO)_6\right]$



(ii) Non- bridged metal carbonyls which contain terminal carbonyls as well as Metal-Metal bonds. For examples,The structure of $Mn_2(CO)_{10}$actually involve only a metal–metal bond, so the formula is more correctly represented as $(CO)_5Mn–Mn(CO)_5$.

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